

Dataset	ε	Nodes	$C_{\max}$	GPU FP32			CPU FP64		
				Time (s)	Δ_{GPU}	Pass	Time (s)	Δ_{CPU}	Pass
GSE24080	Exact degree	27,338	0	1.172 ± 0.035	1.87×10^{-3}	—	—	—	—
	10^{-1}	44–53	7.69×10^{-2}	0.090 ± 0.020	2.03×10^{-3}	5/5	0.156 ± 0.013	1.21×10^{-11}	5/5
	10^{-2}	45–54	7.09×10^{-3}	0.085 ± 0.007	2.15×10^{-3}	5/5	0.164 ± 0.012	1.28×10^{-11}	5/5
	10^{-3}	46–54	8.82×10^{-4}	0.086 ± 0.006	2.15×10^{-3}	1/5	0.162 ± 0.010	1.28×10^{-11}	5/5
TCGA LGG	Exact degree	198,033	0	66.377 ± 1.196	3.32×10^{-2}	—	—	—	—
	10^{-1}	37–49	7.01×10^{-2}	0.320 ± 0.028	3.05×10^{-2}	5/5	0.992 ± 0.076	1.40×10^{-10}	5/5
	10^{-2}	38–50	6.51×10^{-3}	0.312 ± 0.006	3.52×10^{-2}	3/5	1.009 ± 0.097	1.31×10^{-10}	5/5
	10^{-3}	39–51	5.80×10^{-4}	0.310 ± 0.011	3.91×10^{-2}	0/5	1.035 ± 0.100	1.59×10^{-10}	5/5

Table 1: CPU/GPU Cox explanations using 30 fixed training-background patients and five distinct held-out patients per cohort on an Apple M4 Pro. Times are mean $\pm$ sample standard deviation across patients, with one synchronized call per patient and method. CPU FP64 timings are newly measured; GPU FP32 timings reuse the matched saved experiments. Both Δ_{GPU} and Δ_{CPU} are maximum absolute feature-wise errors over all five patients against the same independent, tightly bounded 64-node FP64 reference. These errors differ in definition from the earlier discrepancy against GPU degree-exact output. Pass counts patients whose maximum error is at most ε . $C_{\max}$ is the largest analytical quadrature bound across patients. CPU degree-exact results are unmeasured (—); exact GPU output also contains rounding error. API timings include automatic node selection for approximate rows, transfers, and host accumulation, but exclude model fitting, compilation/warm-up, and cached quadrature-rule construction. Previously measured construction of the exact rules took 10.97s and 561.92s for GSE24080 and TCGA LGG; approximate-rule construction took 0.386–0.563ms. The analytical certificate bounds quadrature error in real arithmetic, not floating-point error. Comparing GPU FP32 with CPU FP64 changes both device and precision; their timing ratio is not a matched-precision GPU speedup.

Separating approximation error from numerical precision. To determine whether the observed discrepancies arise from the quadrature approximation or its numerical evaluation, we compare GPU FP32, CPU FP32, and CPU FP64 explanations for the same fitted Cox models and empirical background games. We retain the 30 fixed training-background patients, five distinct held-out patients per dataset, and automatic node choices for $\varepsilon \in \{10^{-1}, 10^{-2}, 10^{-3}\}$. The GPU results are reused from the saved experiments; both CPU controls are newly measured. Inputs, numerical sources, and package versions were verified against the GPU experiment. The CPU controls evaluate the same parallel product-game formula at the same selected node counts. CPU FP32 provides a control on the same hardware as CPU FP64, so that changing the device is not necessary to investigate the effect of precision.

Across the three tolerances, mean CPU FP64 latencies are 156–164ms for GSE24080 and 992–1,035ms for TCGA LGG, compared with 85–90ms and 310–320ms for the saved GPU FP32 runs. These are measured precision/device tradeoffs, not predictions of FP64 GPU performance.

All three configurations are compared with the same independent 64-node FP64 reference ϕ_p^{ref} :

$$\Delta_a = \max_{p=1,\dots,5} \|\hat{\phi}_{m_p,p}^a - \phi_p^{\text{ref}}\|_{\infty}, \quad a \in \{\text{GPU32}, \text{CPU32}, \text{CPU64}\}.$$

This reference has analytical quadrature bounds below 10^{-10} and a 96-node cross-check for the first patient of each dataset. It is a high-accuracy numerical reference, not a full degree-exact FP64 computation. In particular, Δ_{GPU} in Table 1 now uses this common reference; it is not the discrepancy against rounded GPU degree-exact output reported in the earlier table. All automatically selected rules satisfy their analytical bounds $C_p \leq \varepsilon$ in real arithmetic.

Against the common reference, GPU FP32 meets the requested threshold in 19/30 patient–tolerance cases, CPU FP32 in 19/30, and CPU FP64 in 30/30. The worst errors over these cases are 3.91×10^{-2} , 3.98×10^{-2} , and 1.59×10^{-10} , respectively. The reduction in error from CPU FP32 to CPU FP64, with the

Dataset	ε	CPU FP32 time (s)	Δ_{CPU32}	Pass	CPU32-GPU32 gap
GSE24080	10^{-1}	0.143 ± 0.091	1.99×10^{-3}	5/5	1.79×10^{-4}
GSE24080	10^{-2}	0.103 ± 0.006	2.15×10^{-3}	5/5	9.78×10^{-5}
GSE24080	10^{-3}	0.102 ± 0.004	2.15×10^{-3}	1/5	9.90×10^{-5}
TCGA LGG	10^{-1}	0.619 ± 0.033	3.06×10^{-2}	5/5	2.62×10^{-4}
TCGA LGG	10^{-2}	0.629 ± 0.043	3.56×10^{-2}	3/5	7.17×10^{-4}
TCGA LGG	10^{-3}	0.643 ± 0.048	3.98×10^{-2}	0/5	6.52×10^{-4}

Table 2: Additional CPU FP32 control using the same patients, backgrounds, node counts, formula, and timing protocol. The CPU32-GPU32 gap is the largest absolute difference between their attribution outputs across patients and features. Δ_{CPU32} instead compares CPU FP32 with the same independent FP64 reference used in Table 1. Comparing CPU FP32 and CPU FP64 holds the device and formula fixed while changing work precision; comparing CPU FP32 and GPU FP32 also exposes backend-dependent numerical effects. Both FP32 configurations use FP64 host accumulation.

device and formula held fixed, supports limited work precision as an important source of the discrepancies. Differences between the CPU FP32 and GPU FP32 outputs additionally show the role of backend-specific evaluation. These numerical controls support this explanation but do not constitute a formal floating-point error bound.

For timing, each distinct dataset/node-count configuration is warmed up once before its first measured call. We record one synchronized API call per patient and method, including automatic node selection, data movement, and host accumulation. Model fitting, compilation/warm-up, and cached rule construction are excluded. The CPU and GPU runs use an 8 GiB planning budget and one background row per block; this is a planning budget, not measured memory consumption. Reported standard deviations describe variation across patients rather than repeated-call uncertainty. The saved GPU timings and new CPU timings were recorded in separate runs. The CPU FP64/GPU FP32 latency ratio changes both device and precision, so it should not be interpreted as a matched-precision GPU speedup. Full degree-exact CPU runtimes were not measured and are left blank.

The analytical guarantee remains a bound on quadrature error for the fixed fitted model and chosen 30-row empirical background game. It excludes floating-point rounding, uncertainty in the fitted model, and error from replacing a population background with this empirical background. Small FP64 discrepancies are an empirical accuracy check, not a stronger analytical certificate. The observed accuracy of CPU FP64 must therefore be paired with its CPU runtime rather than with the faster GPU FP32 runtime.